

AI-HRMS: AI-Powered Human Resource Management System

Module-wise Functionality Document

A comprehensive document for the AI-Powered Human Resource Management System, covering module specifications, operational procedures, and system architecture.

Classification: Internal - Confidential

Date: March 2025

Organization: AI-HRMS Product Team

Table of Contents

1. Document Purpose and Scope	2
2. System Overview	2
3. Role Definitions and Hierarchy	2
4. Module: Dashboard	3
5. Module: Employee Management	4
6. Module: RBAC and Security	5
7. Module: AI Talent Acquisition	6
8. Module: Time and Attendance	7
9. Module: Payroll and Expenses	8
10. Module: Performance and Talent	8
11. Module: Learning and Development	9
12. Module: Analytics and Reporting	10
13. Module: Employee Self-Service	11
14. Configured Workflows	12
15. Complete RBAC Permission Matrix	13
16. Workflow State Machine Definitions	13
17. Notification Triggers	14
18. Cross-Module Integration Map	14
19. Data Flow Diagrams	15
20. Appendix	15

1. Document Purpose and Scope

This document provides a comprehensive, module-wise functional specification for the AI-powered Human Resource Management System (HRMS). It serves as the authoritative reference for Development Teams, QA Teams, Product Stakeholders, and Operations Teams.

This document covers ten functional modules, seven organizational roles, nine configured workflows, cross-module integration specifications, and AI feature definitions. It does not cover infrastructure, deployment topology, or non-functional requirements.

2. System Overview

The AI-powered HRMS is a unified, cloud-native platform designed to digitize and automate the full spectrum of human resource operations - from talent acquisition through employee lifecycle management to offboarding. The system leverages Artificial Intelligence across modules for predictive analytics, intelligent screening, automated recommendations, and anomaly detection.

Architectural Principles

Principle	Description
Role-Driven Access	Every feature is gated by a hierarchical RBAC model with Level 0-4 granularity
Workflow-First Design	All state-changing operations traverse configurable approval workflows
AI-Augmented Operations	AI assists but does not autonomously execute critical decisions without human confirmation
Audit Trail Completeness	Every CREATE, UPDATE, DELETE operation is logged with actor, timestamp, and delta
Module Independence	Modules are independently deployable but communicate via event bus for cross-cutting concerns

3. Role Definitions and Hierarchy

Role	Level	Scope of Authority	Delegation
Super Admin	0	Unlimited - full system configuration, all modules, all data	Yes
HR Admin	1	HR operations, payroll oversight, performance, attendance, lifecycle	Yes (HR)
Payroll Specialist	2	Payroll processing, tax declarations, statutory compliance	No

Role	Level	Scope of Authority	Delegation
Department Manager	3	Team management, leave/expense approvals, performance reviews	Yes (acting)
Employee	4	Self-service only - view own data, submit requests	No
Recruiter	2	Job postings, candidate pipeline, interview scheduling, AI screening	No
L&D; Manager	2	Course catalog, skill assessments, training program management	No

Role Hierarchy Inheritance Rules

1. Super Admin inherits all permissions of every role.
2. HR Admin inherits Employee-level self-service permissions.
3. Department Manager inherits Employee-level self-service permissions.
4. Roles at the same level (Level 2) do not inherit from each other - Payroll Specialist, Recruiter, and L&D; Manager have orthogonal permission sets.
5. Delegation is time-bound (max 30 days) and auditable.

4. Module: Dashboard

4.1 Module Overview and Purpose

The Dashboard module central command center for all HRMS users.

4.2 Feature List

1. Role-Adaptive Widgets
2. KPI Scorecards
3. Pending Actions Panel
4. AI Insight Cards
5. Quick Action Buttons
6. Announcements Banner
7. Calendar Integration
8. Trend Charts
9. Document Expiry Alerts
10. Team Availability View

4.3 AI Features

AI Feature	Input	Processing	Output
Predictive Attrition Alert	Employee tenure, engagement scores, salary benchmark	Gradient-boosted classification model with 90-day prediction window	Risk score (0-100), contributing factors, retention actions
Hiring Demand Forecast	Historical attrition, open requisitions, growth rate	Time-series forecasting (ARIMA + ensemble)	30/60/90-day hiring demand by department
Engagement Score Predictor	Pulse survey responses, attendance, LMS activity	NLP sentiment analysis + behavioral clustering	Engagement score (1-10), trend direction, at-risk list

5. Module: Employee Management

5.1 Module Overview and Purpose

The Employee Management module system of record for all employee lifecycle data.

5.2 Feature List

1. Employee Master Data
2. Employment Records
3. Organization Chart
4. Document Management
5. Asset Assignment
6. Job History Timeline
7. Onboarding Checklist
8. Offboarding Workflow
9. Bulk Import/Export
10. AI Duplicate Detection
11. Employee Directory
12. Probation Management

5.3 AI Features

AI Feature	Input	Processing	Output
Duplicate Record Detection	Employee name, DOB, email, phone, national ID	Fuzzy matching with Levenshtein distance + phonetic matching	Potential duplicate pairs with confidence score

AI Feature	Input	Processing	Output
Onboarding Task Prediction	Role, department, location, access requirements	ML model trained on historical onboarding patterns	Auto-generated onboarding checklist
Flight Risk Assessment	Tenure, compensation benchmark, engagement, manager rating	Ensemble model (logistic regression + XGBoost)	Risk classification with retention strategy

6. Module: RBAC and Security

6.1 Module Overview and Purpose

The RBAC and Security module enforces the principle of least privilege.

6.2 Feature List

1. User Account Management
2. Role Assignment Engine
3. Permission Policy Editor
4. MFA
5. Session Management
6. Audit Log Viewer
7. Data Encryption Management
8. IP Whitelisting
9. Delegation Management
10. Compliance Reporting
11. Login Analytics
12. API Key Management

6.3 AI Features

AI Feature	Input	Processing	Output
Anomalous Login Detection	Login time, IP geolocation, device fingerprint	Isolation Forest anomaly detection	Risk score, suggested action, incident ticket
Privilege Escalation Detection	Role change history, access patterns, permission utilization	Behavioral analysis: actual usage vs. granted permissions	Unused permission report, over-provisioning alerts

AI Feature	Input	Processing	Output
Insider Threat Scoring	Data access patterns, bulk downloads, off-hours activity	Graph-based anomaly detection across user behavior network	Threat score, flagged activities, investigation actions

7. Module: AI Talent Acquisition

7.1 Module Overview and Purpose

The AI Talent Acquisition module automates and optimizes end-to-end recruitment.

7.2 Feature List

1. Job Requisition Management
2. Job Posting Multi-Channel
3. AI Resume Screening
4. Candidate Pipeline (Kanban)
5. Interview Scheduling
6. Assessment Integration
7. Offer Letter Generation
8. AI Bias Detection
9. Candidate Communication
10. Referral Management
11. Recruitment Analytics
12. AI Candidate Matching
13. Talent Pool Management

7.3 AI Features

AI Feature	Input	Processing	Output
Resume Parser and Screener	Resume file (PDF/DOCX), job description, required skills	NLP-based extraction (spaCy + custom NER)	Structured profile, match score (0-100), ranking
Bias Detection Engine	Job description text, screening criteria, historical data	NLP with debiased word embeddings, statistical parity	Bias score per category, suggested rewording
Candidate-Job Matching	Candidate profile, job requirements, team composition	Multi-dimensional matching: skills, experience, culture fit	Match percentage, ranked list, hiring recommendation

8. Module: Time and Attendance

8.1 Module Overview and Purpose

The Time and Attendance module captures, validates, and processes work-time data.

8.2 Feature List

1. Check-In/Check-Out
2. Shift Management
3. Leave Management
4. Holiday Calendar
5. Overtime Tracking
6. Attendance Regularization
7. Comp-Off Management
8. Timesheet Management
9. AI Anomaly Detection
10. Attendance Policy Engine
11. Leave Encashment
12. WFH Tracking

8.3 AI Features

AI Feature	Input	Processing	Output
Buddy Punching Detection	Check-in/out timestamps, geolocation, device IDs	Clustering analysis on co-located check-ins	Suspicious pairs report, confidence score
Absenteeism Prediction	Historical leave patterns, engagement, workload	LSTM-based time-series model with attention	30-day absence probability, high-risk group ID
Optimal Shift Scheduling	Employee preferences, workload, labor laws, skill coverage	Constraint satisfaction optimization (genetic algorithm)	Proposed shift roster, coverage analysis

9. Module: Payroll and Expenses

9.1 Module Overview and Purpose

The Payroll and Expenses module handles complete payroll lifecycle.

9.2 Feature List

- 1. Salary Structure Definition
- 2. Payroll Processing
- 3. Tax Calculation Engine
- 4. Statutory Compliance
- 5. Expense Claim Management
- 6. Advance Salary
- 7. Loan Management
- 8. Payslip Generation
- 9. Bank Integration
- 10. Tax Declaration
- 11. AI Payroll Anomaly Detection
- 12. Year-End Processing
- 13. Reimbursement Policies

9.3 AI Features

AI Feature	Input	Processing	Output
Payroll Anomaly Detection	Payroll register, historical patterns, organizational averages	Statistical outlier detection (Z-score + IQR)	Flagged entries with anomaly type, severity
Expense Fraud Detection	Expense claims, receipt images, historical claims	OCR on receipts, duplicate image detection, pattern analysis	Fraud probability score, flagged claims
Salary Benchmarking AI	Role, experience, location, market data	Regression analysis on multi-source market data	Suggested salary band, market percentile

10. Module: Performance and Talent

10.1 Module Overview and Purpose

The Performance and Talent module manages performance evaluation lifecycle.

10.2 Feature List

- 1. Goal Setting (OKR/KPI)
- 2. Review Cycle Management
- 3. Self-Assessment
- 4. Peer Review (360)
- 5. Manager Review
- 6. Calibration Sessions
- 7. PIP
- 8. Succession Planning
- 9. 9-Box Talent Matrix
- 10. Competency Framework
- 11. AI Performance Insights
- 12. Promotion Workflow
- 13. Skill Gap Analysis

10.3 AI Features

AI Feature	Input	Processing	Output
Feedback Sentiment Analysis	Written feedback text from all review channels	NLP sentiment analysis (BERT-based), emotion detection	Sentiment score, key themes, flagged feedback
Rating Bias Detector	Ratings by manager, department, demographics	Statistical parity analysis, regression for attribute correlation	Bias indicators, rating adjustment recommendations
Succession Readiness Score	Performance history, skill assessments, leadership potential	Multi-factor scoring model with weighted criteria	Readiness score (1-100), development gap analysis

11. Module: Learning and Development

11.1 Module Overview and Purpose

The Learning and Development module manages course catalog, learning paths, and skill inventory.

11.2 Feature List

- 1. Course Catalog

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2. Learning Paths
 3. Skill Inventory
 4. AI Training Recommendations
 5. Assessment Engine
 6. Certification Tracking
 7. Competency Mapping
 8. Mandatory Training Assignment
 9. Learning Analytics
 10. Content Management

11.3 AI Features

AI Feature	Input	Processing	Output
Course Recommendation Engine	Employee skill profile, skill gaps, career goals, role requirements	Collaborative filtering + content-based filtering hybrid	Personalized course recommendations ranked by relevance
Skill Gap Auto-Detection	Role competency model, current skill assessments, training history	ML model comparing actual vs. required proficiency	Skill gap report with priority ranking and training suggestions
Learning Path Optimization	Employee learning style, available time, career trajectory	Reinforcement learning on learning outcome data	Optimized learning path sequence and pacing

12. Module: Analytics and Reporting

12.1 Module Overview and Purpose

The Analytics and Reporting module provides dashboards, predictive analytics, and custom reports.

12.2 Feature List

1. Real-Time Dashboards (6 types)
2. Predictive Analytics (4 models)
3. Custom Report Builder
4. Data Export (PDF/Excel/CSV)
5. Department-Specific Views
6. KPI Tracking
7. Trend Analysis
8. Benchmarking

12.3 AI Features

AI Feature	Input	Processing	Output
Attrition Risk Prediction	Employee demographics, engagement, performance, market data	Gradient-boosted classification with 90-day window	Risk score, contributing factors, recommended actions
Headcount Forecast	Historical headcount, business growth rate, seasonal patterns	Time-series forecasting with confidence intervals	Projected headcount by department for 6/12 months
Training Completion Forecast	Enrollment data, course difficulty, historical completion rates	Survival analysis on enrollment data	Predicted completion dates, at-risk enrollments

13. Module: Employee Self-Service

13.1 Module Overview and Purpose

The Employee Self-Service module centralized portal for employee self-service.

13.2 Feature List

1. Profile Management
2. Leave Requests
3. Expense Submission
4. Payslip Access
5. AI Chat Assistant
6. Document Access
7. Training Enrollment
8. Benefit Selection
9. Feedback Submission
10. Directory Search

13.3 AI Features

AI Feature	Input	Processing	Output
AI HR Assistant	Natural language queries about HR policies, procedures, and personal data	LLM-based conversational AI with HR domain fine-tuning	Context-aware responses with reference links and actionable suggestions
Smart Form Completion	Partial form data, employee profile, historical patterns	Auto-fill based on profile data and learned patterns	Pre-populated form fields with suggestions
Proactive Notification	Employee data changes, policy updates, deadline proximity	Rule engine + ML prediction of relevant notifications	Timely, personalized notifications for required actions

14. Configured Workflows

The HRMS platform implements nine configured workflows that govern state-changing operations across modules. Each workflow follows a multi-level approval pattern with configurable escalation rules and SLA tracking.

Workflow	Trigger	Approval Levels	SLA
Leave Approval	Employee submits leave request	Manager → HR (if > 3 days)	24 hours
Expense Approval	Employee submits expense claim	Manager → Finance	48 hours
Payroll Processing	HR Admin initiates monthly payroll	HR Admin → Finance verification	End of month
Performance Review	Review cycle published	Self → Peer → Manager → Calibration	Cycle duration
Onboarding	Candidate marked as "Hired"	IT → HR → Manager → L&D;	5 business days
Offboarding	Employee status set to "Inactive"	IT → Finance → HR clearance	10 business days
Job Requisition	Department Manager creates requisition	HR Admin → Budget approval	3 business days
Promotion Nomination	Manager nominates employee	HR review → Committee approval	15 business days
Policy Update	HR Admin creates/updates policy	Legal review → HR Admin approval	5 business days

15. Complete RBAC Permission Matrix

The following matrix summarizes the permission levels across all modules for each role. Permission levels: ADMIN = full control, WRITE = create/edit, READ = view only, - = no access.

Module	Super Admin	HR Admin	Payroll	Dept Mgr	Employee	Recruiter	L&D; Mgr
Dashboard	ADMIN	ADMIN	READ	READ	READ(own)	READ	READ
Employees	ADMIN	WRITE	READ	READ(team)	READ(own)	READ	READ
RBAC	ADMIN	WRITE	-	-	-	-	-
Talent	ADMIN	WRITE	-	READ(team)	-	ADMIN	-
Attendance	ADMIN	WRITE	READ	WRITE(approve)	WRITE(own)	WRITE(own)	WRITE(own)
Payroll	ADMIN	WRITE(approve)	ADMIN	-	READ(own)	-	-
Performance	ADMIN	WRITE	-	WRITE(team)	WRITE(own)	-	READ
Learning	ADMIN	WRITE	-	-	WRITE(enroll)	-	ADMIN
Analytics	ADMIN	ADMIN	READ(payload)	READ(team)	-	READ	READ
Self-Service	ADMIN	ADMIN	WRITE(own)	WRITE(own)	WRITE(own)	WRITE(own)	WRITE(own)

16. Workflow State Machine Definitions

Workflow	States	Transitions
Leave	Pending → Approved/Rejected	Manager approves/rejects; HR overrides for > 3 days
Expense	Pending → Manager Approved → Finance Approved → Reimbursed / Rejected	Each level approves/rejects; rejected = terminal
Payroll	Draft → Processing → Verified → Paid	HR initiates, finance verifies, system disburses
Candidate	Applied → Screening → Interview → Offered → Hired / Rejected	Recruiter advances/rejects at each stage
Onboarding	Pending → In Progress → Completed	Checklist items auto-transition on completion
Performance	Draft → In Review → Completed	Self → Manager → Calibration sequence

17. Notification Triggers

Trigger Event	Recipients	Channel	Priority
Leave request submitted	Reporting Manager	Email + In-app	Normal
Leave approved/rejected	Employee	Email + In-app	Normal
Expense claim approved	Employee, Finance	In-app	Normal
Payroll processed	All employees	Email	High
AI attrition alert (high risk)	HR Admin, Department Manager	Email + In-app	Critical
Document expiry warning (30/15/7 days)	Employee, HR Admin	Email + In-app	High
Review cycle published	All participants	Email + In-app	Normal
Onboarding task overdue	HR Admin, Manager, New Hire	Email + In-app	High
Policy updated	All employees	Email + In-app	Normal
Suspicious login detected	User, Super Admin	Email + SMS	Critical

18. Cross-Module Integration Map

Source Module	Target Module	Data Flow	Integration Type
Employee Management	RBAC & Security	New employee → user account creation	Write
Employee Management	Payroll & Expenses	Bank details, salary structure	Write
Time & Attendance	Payroll & Expenses	Attendance data, leave deductions, OT hours	Write
AI Talent Acquisition	Employee Management	Hired candidate → new employee record	Write
Performance & Talent	Learning & Development	Skill gaps → mandatory training assignments	Write
All Modules	Dashboard	Aggregated metrics and KPIs	Read
All Modules	Analytics & Reporting	Data for dashboards and custom reports	Read
All Modules	Audit Log	CRUD operations logged with actor and delta	Write

Source Module	Target Module	Data Flow	Integration Type
RBAC & Security	All Modules	Permission checks on every API request	Enforce

19. Data Flow Diagrams

The data flow architecture follows a hub-and-spoke model with Employee Management as the central hub. Employee data feeds into all other modules, while cross-module communication is event-driven. Key data flows include: Employee → Payroll (salary structure), Attendance → Payroll (time records), Talent Acquisition → Employee (new hire creation), Performance → Learning (skill gap training), and all modules → Audit Log (operation logging).

20. Appendix

A. Document Revision History

Version	Date	Author	Changes
1.0.0	2025-01-15	HRMS Product Team	Initial release
2.0.0	2025-03-05	HRMS Product Team	Added AI features, RBAC matrices, workflows

B. Acronyms

Acronym	Full Form
HRMS	Human Resource Management System
RBAC	Role-Based Access Control
OKR	Objectives and Key Results
MFA	Multi-Factor Authentication
LSTM	Long Short-Term Memory
NER	Named Entity Recognition
NLP	Natural Language Processing
KPI	Key Performance Indicator
CTC	Cost to Company
PIP	Performance Improvement Plan